

SION AP, Switzerland

WMO: 067200

Lat: 46.2186N

Lon: 7.3303E

Elev: 1581

StdP: 13.88

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.2	(c) 20.3	(d) 10.3	(e) 9.9	(f) 24.1	(g) 13.4	(h) 11.5	(i) 25.8	(j) 18.2	(k) 41.2	(l) 15.2	(m) 40.5	(n) 3.3	(o) 80	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 89.1	(d) 67.7	(e) 85.6	(f) 66.6	(g) 82.6	(h) 65.4	(i) 69.1	(j) 85.5	(k) 67.7	(l) 82.7	(m) 66.4	(n) 80.0	(o) 11.3	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.8	(b) 94.0	(c) 72.9	(d) 62.5	(e) 89.7	(f) 72.3	(g) 61.3	(h) 85.9	(i) 71.9	(j) 34.1	(k) 85.8	(l) 32.9	(m) 83.0	(n) 31.8	(o) 80.3	(p) 73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.2	(b) 17.9	(c) 15.5	(e) 14.0	(f) 94.1	(g) 4.3	(h) 3.6	(i) 10.9	(j) 96.8	(k) 8.4	(l) 98.9	(m) 5.9	(n) 100.9	(o) 2.8	(p) 103.6		
(4) 20.2	(5) 17.9	(6) 15.5	(7) 13.1	(8) 71.8	(9) 4.2	(10) 1.2	(11) 10.1	(12) 72.7	(13) 7.7	(14) 73.4	(15) 5.3	(16) 74.0	(17) 2.3	(18) 74.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	33.9	37.5	45.9	52.9	60.1	67.0	69.9	68.5	61.2	53.2	42.2	34.6		
	DBStd	14.09	6.13	6.53	6.39	6.21	5.73	6.12	5.50	5.52	5.52	5.77	6.84	6.07		
	HDD50	1807	500	349	156	41	3	0	0	0	1	34	245	478		
	HDD65	5087	965	769	591	364	170	49	20	28	139	365	684	943		
	CDD50	2654	0	0	30	127	318	510	616	572	336	134	11	0		
	CDD65	460	0	0	0	1	19	110	170	135	25	0	0	0		
	CDH74	4150	0	0	0	38	244	1037	1517	1123	187	4	0	0		
	CDH80	1283	0	0	0	2	40	329	519	369	24	0	0	0		
(14) Wind	WSAvg	4.8	3.3	4.0	5.3	6.1	6.1	6.3	6.1	5.3	5.0	3.8	3.3	3.2		
(15) Precipitation	PrecAvg	32.2	2.4	2.4	2.1	2.3	3.1	3.0	3.3	3.1	2.4	2.4	2.8	2.9		
	PrecMax	43.8	7.3	9.0	6.0	4.3	6.1	5.7	6.7	6.1	5.7	6.1	7.4	8.1		
	PrecMin	23.4	0.4	0.2	0.8	0.5	0.8	1.4	1.6	1.1	0.7	0.2	0.2	0.0		
	PrecStd	4.9	1.8	2.1	1.3	1.1	1.7	1.2	1.2	1.3	1.2	1.4	1.7	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	59.7	69.8	78.7	84.3	91.7	93.3	93.3	83.4	74.2	63.8	53.9		
		MCWB	44.0	45.8	51.3	57.2	63.5	68.5	68.5	69.2	64.8	58.3	51.4	45.2		
	2%	DB	48.1	55.0	65.9	73.6	80.0	87.4	89.2	88.0	78.9	70.0	59.4	48.4		
		MCWB	41.4	43.3	50.0	54.9	61.2	67.1	67.6	67.7	63.3	57.3	49.6	43.1		
	5%	DB	44.7	51.0	61.9	69.9	76.4	83.9	85.5	83.9	75.6	66.6	55.4	45.3		
		MCWB	40.0	41.6	48.2	53.2	59.4	65.5	66.7	66.5	62.1	55.7	47.7	41.4		
	10%	DB	42.1	47.2	57.8	65.9	72.8	80.2	82.3	80.4	72.2	63.1	51.9	43.1		
		MCWB	38.1	40.2	46.2	51.3	57.7	63.8	65.6	65.2	60.2	54.4	46.4	39.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	47.4	53.1	58.4	65.1	70.5	70.9	70.8	66.6	61.2	53.0	46.6		
		MCDB	50.4	55.7	66.4	75.3	80.9	88.3	88.1	88.2	78.7	68.5	59.6	51.3		
	2%	WB	42.8	44.7	51.0	55.9	62.5	68.3	69.1	68.9	64.6	59.2	50.7	44.1		
		MCDB	46.3	52.8	63.1	71.0	77.0	84.6	85.6	84.8	76.2	66.3	56.7	47.2		
	5%	WB	40.7	42.6	49.0	54.0	60.5	66.5	67.6	67.3	63.0	57.3	48.9	42.3		
		MCDB	44.0	49.5	59.6	67.5	73.4	81.1	82.9	81.2	73.4	64.0	54.3	45.0		
	10%	WB	38.7	40.5	47.2	52.2	58.8	64.8	66.3	65.9	61.3	55.5	46.9	40.1		
		MCDB	41.6	46.5	56.3	64.4	70.3	77.8	80.3	78.1	70.4	61.9	51.5	42.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	16.2	20.0	21.0	20.8	21.2	21.5	20.6	19.4	18.4	15.0	12.1		
		MCDBR	15.2	22.3	26.7	27.6	27.1	26.7	26.8	26.7	23.6	22.1	19.3	14.3		
		MCWBR	9.9	12.6	13.7	12.7	12.0	11.0	10.3	10.9	10.8	11.4	11.0	9.5		
	5% WB	MCDBR	13.5	20.1	24.2	25.5	24.3	24.3	24.5	23.9	21.3	19.3	16.8	12.5		
		MCWBR	9.4	12.1	12.8	12.3	11.4	10.6	10.2	10.4	10.1	10.5	10.1	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.300	0.352	0.398	0.419	0.423	0.419	0.405	0.378	0.350	0.307	0.271		
	taud		2.305	2.231	2.149	2.136	2.174	2.224	2.234	2.285	2.343	2.392	2.402	2.347		
	Ebn at Noon		263	275	274	271	269	268	267	267	264	255	247	250		
	Edh at Noon		24	32	41	45	45	43	43	39	34	28	22	21		
(44) All-Sky Solar Radiation	RadAvg		462	738	1090	1391	1587	1888	1882	1640	1284	833	477	379		
	RadStd		52	101	149	188	135	168	154	135	107	89	67	56		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.94	N/A	N/A	+2.39	+0.52	N/A	N/A	+261	+130
(47) Regional (105 neighbors)		N/A	N/A	N/A	+1.28	+0.68	N/A	N/A	+99	+31

Nomenclature: See separate page